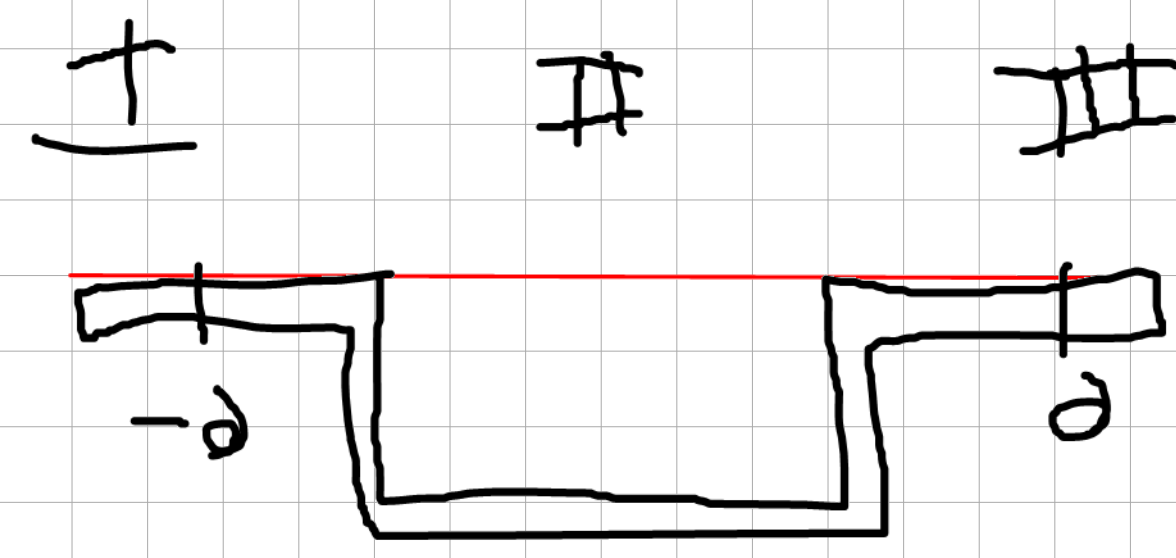


Quantum Mek I 5. Bölüm 2. Kısım

Tekrar

$$\begin{aligned} e^{-ikx} + R e^{ikx} &= A e^{-iqx} + B e^{iqx} \\ ik(-ik) - ik(ik) &= iq(A e^{-iqx} - B e^{iqx}) \quad iq(A e^{-iqx} - B e^{iqx}) = ik T e^{ikx} \end{aligned}$$



R ve T nedir?

$$\begin{aligned} \text{(ii) için;} \quad A e^{iqx} + B e^{-iqx} &= T e^{ikx} \\ \frac{k}{q} (A e^{iqx} - B e^{-iqx}) &= T e^{ikx} \end{aligned} \quad \begin{aligned} (+) \quad \frac{q+k}{q} (A e^{iqx}) + \frac{q-k}{q} (B e^{-iqx}) &= 2 T e^{ikx} \\ (-) \quad \frac{q-k}{q} (A e^{iqx}) + \frac{q+k}{q} (B e^{-iqx}) &= 0 \end{aligned}$$

$$\begin{aligned} \text{(i) için;} \quad e^{-ikx} + R e^{ikx} &= A e^{-iqx} + B e^{iqx} \\ e^{-ikx} - R e^{ikx} &= \frac{q}{k} (A e^{-iqx} - B e^{iqx}) \end{aligned} \quad \begin{aligned} (+) \quad 2 e^{-ikx} &= \frac{k+q}{k} (A e^{-iqx}) + \frac{k-q}{k} (B e^{iqx}) \\ (-) \quad 2 R e^{ikx} &= \frac{k-q}{k} (A e^{-iqx}) + \frac{k+q}{k} (B e^{iqx}) \end{aligned}$$

$$\begin{aligned} \text{ii için;} \quad A e^{iqx} + B e^{-iqx} &= T e^{ikx} \\ A e^{iqx} - B e^{-iqx} &= \frac{k}{q} T e^{ikx} \\ (+) \quad 2 A e^{iqx} &= \frac{k+q}{q} T e^{ikx} \\ (-) \quad 2 B e^{-iqx} &= \frac{q-k}{q} T e^{ikx} \end{aligned}$$

$$\text{ii için;} \quad \left(\frac{q+k}{q}\right) e^{-ikx} + \left(\frac{q-k}{q}\right) R e^{ikx} = 2 A e^{-iqx} \quad \left(\frac{q-k}{q}\right) e^{-ikx} + \left(\frac{q+k}{q}\right) R e^{ikx} = 2 B e^{iqx}$$

$$\Rightarrow \star A = \frac{k+q}{2q} T e^{i\partial(k-q)} \quad \star B = \frac{q-k}{2q} T e^{i\partial(k+q)}$$

$$\Rightarrow \left(\frac{q+k}{q}\right) e^{-ikx} + \left(\frac{q-k}{q}\right) R e^{ikx} = \frac{k+q}{q} T e^{i\partial(k-2q)} \quad \left(\frac{q-k}{q}\right) e^{-ikx} + \left(\frac{q+k}{q}\right) R e^{ikx} = \frac{q-k}{q} T e^{i\partial(k+2q)}$$

$$T = \frac{q+k}{q} \frac{q}{k+q} e^{-ikx} e^{-i\partial(k-2q)} + \frac{q-k}{q} \frac{q}{k+q} R e^{ikx} e^{-i\partial(k-2q)}$$

$$\Rightarrow \star T = e^{-2i\partial(k+q)} + \frac{q-k}{q+k} R e^{2i\partial q}$$

$$R = \frac{k-q}{2k} A e^{-i\partial(k+q)} + \frac{k+q}{2k} B e^{i\partial(q-k)}$$

$$\Rightarrow R = \frac{k-q}{2k} \left(\frac{k+q}{2q}\right) T e^{i\partial(k-q)} e^{-i\partial(k+q)} + \frac{k+q}{2k} \left(\frac{q-k}{2q}\right) T e^{i\partial(k+q)} e^{i\partial(q-k)}$$

$$\Rightarrow \star \frac{k^2 - q^2}{4kq} T e^{-2i\partial q} + \frac{q^2 - k^2}{4kq} T e^{2i\partial q} = R \Rightarrow R = \frac{k^2 - q^2}{2kq}$$

$$T = e^{-2i\partial(k+q)} + \frac{q-k}{q+k} R e^{2i\partial q} \Rightarrow e^{i\theta} = \cos\theta + i\sin\theta \quad e^{-i\theta} = \cos\theta - i\sin\theta \quad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\Rightarrow \frac{k^2 - q^2}{4kq} T e^{-2i\vartheta} + \frac{q^2 - k^2}{4kq} T e^{2i\vartheta} = R \Rightarrow \frac{T(k^2 - q^2)}{4kq} \left(\frac{e^{-2i\vartheta} - e^{2i\vartheta}}{2} \right)$$

$$\frac{T(q^2 - k^2)}{2kq} \left(\frac{e^{2i\vartheta} - e^{-2i\vartheta}}{2} \right)$$

$$R = i \sin 2\vartheta \cdot T \frac{(q^2 - k^2)}{2kq}$$

$$T = \frac{e^{-2i\vartheta(k+q)}}{e} + \frac{q-k}{q+k} e^{2i\vartheta q}$$

$$T = e^{-2i\vartheta(k+q)} + \frac{q-k}{q+k} \cdot \frac{q-k}{q+k} \frac{T}{2kq} i \sin 2\vartheta \Rightarrow T \left(1 - \frac{(q-k)^2 i \sin 2\vartheta}{2kq} \right) = e^{-2i\vartheta(k+q)}$$

$$\Rightarrow \frac{T}{2kq} (2kq + i(k-q) \sin 2\vartheta) = e^{-2i\vartheta(k+q)}$$

$$\Rightarrow T = \frac{e^{-2i\vartheta(k+q)} \cdot 2kq}{2kq + i(k-q) \sin 2\vartheta}$$

Hata olabilir bir yerde
size bakarsanız

